

Week 3
Inverse Kinematics

ENN581 Robot Motion, Control and Planning

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ACKNOWLEDGEMENT OF TRADITIONAL OWNERS

QUT acknowledges the Turrbal and Yugara, as the First Nations owners of the lands where QUT now stands. We pay respect to their Elders, lores, customs and creation spirits. We recognise that these lands have always been places of teaching, research and learning.

QUT acknowledges the important role Aboriginal and Torres Strait Islander people play within the QUT community.

Last week, we learnt

- The goal of FK is to be able to describe the pose of the end effector with respect to a known base frame.

- There are several methods to assign frames to the links of a serial manipulator.
- No matter the method used, the transformation matrix (or other ways to represent pose) must completely define the configuration of the manipulator arm.

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Questions?

- Is everyone comfortable with the ETS method?
- Have you tried applying it to popular robot arms?
  Franka, ABB, Kuka, PUMA, SCARA, etc?
- Any examples of DH tables that you came across?
- Have you played with CoppeliaSim? Any other simulators that you tried?

- Assessment related questions – at the end of the lecture.

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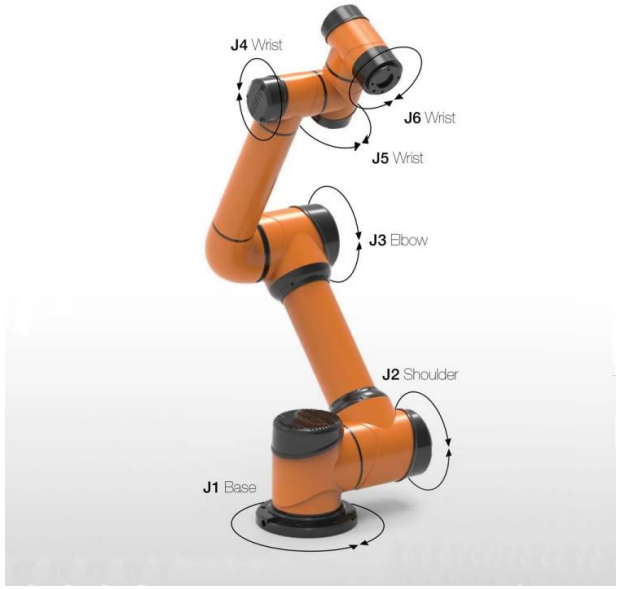

ABB
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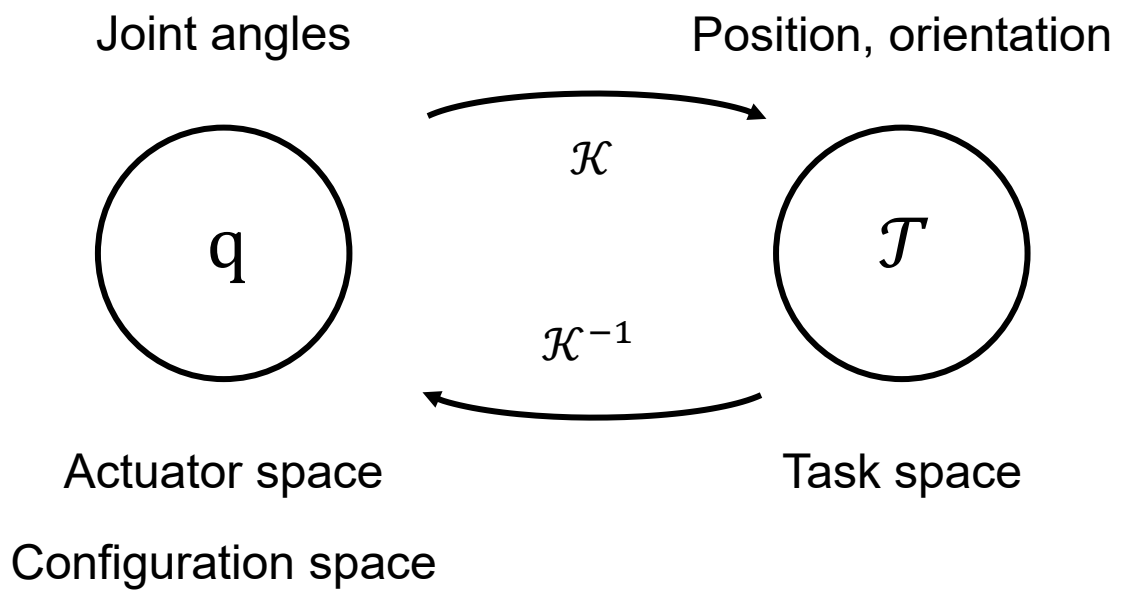
Today's lecture

- Common terms in the description of a robot's configuration
  - Task space
  - Configuration space
- Inverse kinematics
  - What is the formal problem?
  - What are common approaches to the solution?

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Configuration space and Task space





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Configuration space

A robot's configuration can be completely described by a set of generalised coordinates

The set of all possible configurations is called the configuration space: $\mathcal{C}$

E.g., for a manipulator arm with 2 joints,

- The complete configuration can be given in terms of joint configuration $\mathbf{q} = \{\mathrm{q}_1, \mathrm{q}_2\}$
- Here, $\mathrm{q}_i \in \mathbb{S}^1$, where $\mathbb{S}^1 \coloneqq \{s \mid s \in [0, 2\pi)$ is a set of angles
- The configuration space is $\mathcal{C} \subset \mathbb{S}^1 \times \mathbb{S}^1$

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Task space

The space (or set) of all possible end effector poses is the task space: $\mathcal{T}$

$$\xi_n \in \mathcal{T}$$

A general symbol for pose
of end effector
relative to a reference

Could be ${}^0\mathbf{T}_n$, {x,y,z,R,P,Y} , {x, y, z, quaternion}, etc.

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The goal of forward kinematics is

- To give the Cartesian coordinates and orientation of the end effector of a manipulator in terms of joint coordinates

$$\xi_n = \mathcal{K}(\mathbf{q}),\ \mathbf{q} = \{\mathrm{q}_i, i \in [1, \cdots, n]\}$$

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The inverse kinematics problem

Given a desired position and orientation of the end effector, find the joint coordinates q_i, that result in this configuration

$$\mathbf{q} = \mathcal{K}^{-1}(\xi_n)$$

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Key questions to consider when solving the IK problem

- Does a solution exist?
- If yes, how many solutions?
- What is the method to solving the problem?

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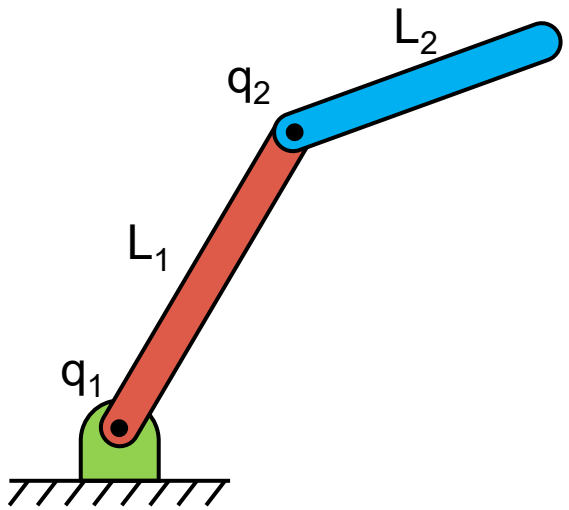
Does a solution exist?

- Recall our discussion about the manipulator's workspace from week 1
- Reachable: the region that end-effector can reach in one orientation
- Dexterous: the region that the end-effector can reach in all its possible orientations

For a solution to exist, the specified goal point must lie within the workspace

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Consider this planar manipulator



Manipulator geometry:
$L_1 \neq L_2$. Say, $L_1 = 1$ m and $L_2 = 0.7$ m.
$q_1 \in [0, 2\pi)$ and $q_2 \in [0, 2\pi)$.
Ignore collisions.

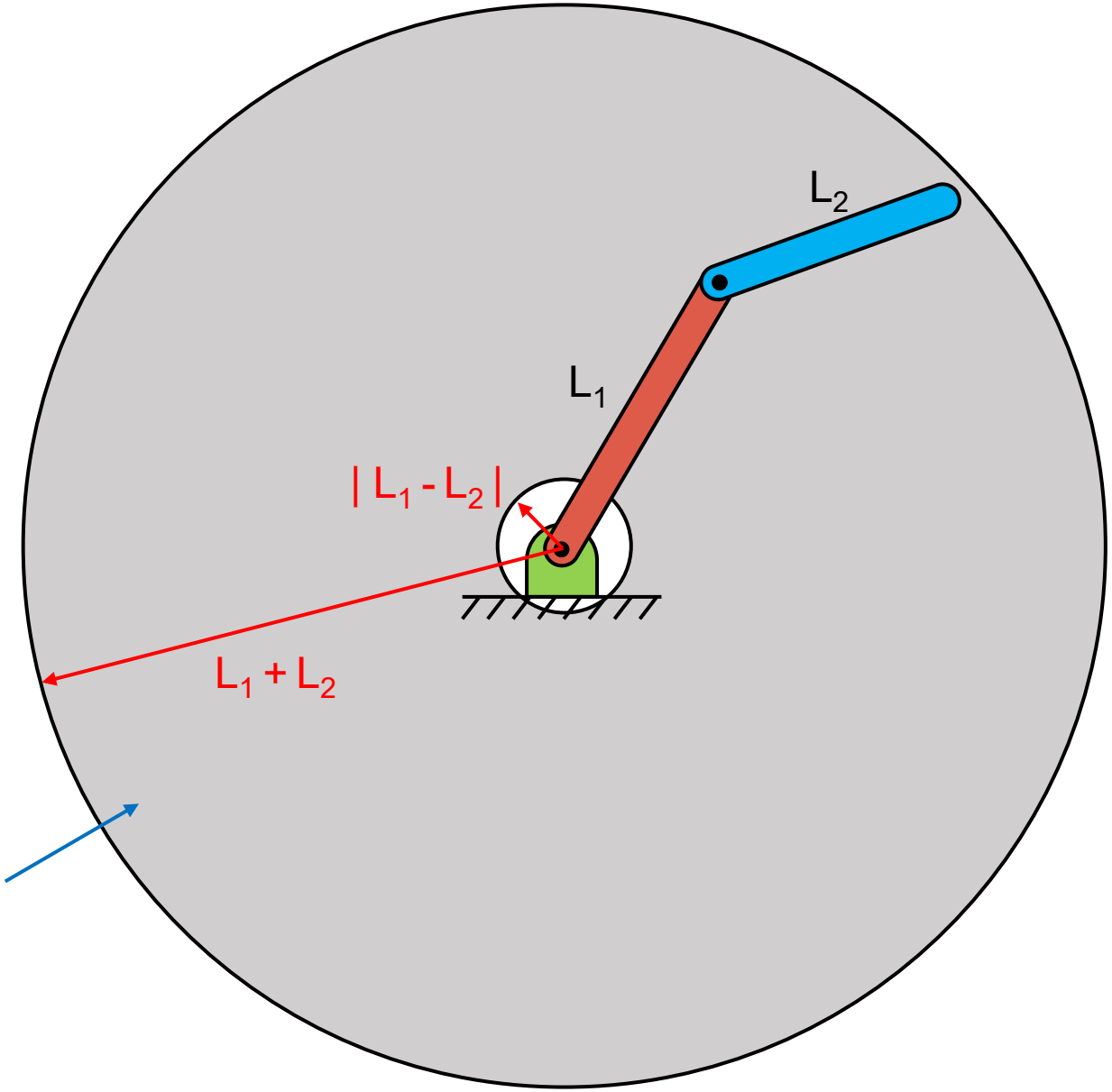
What is the dexterous workspace of this manipulator?

What is the reachable workspace of this manipulator?

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Workspace

$L_1 \neq L_2$. Say L_1 = 1 m, L_2 = 0.7 m.
$q_1 \in [0, 2\pi)$
$q_2 \in [0, 2\pi)$
Ignore collisions



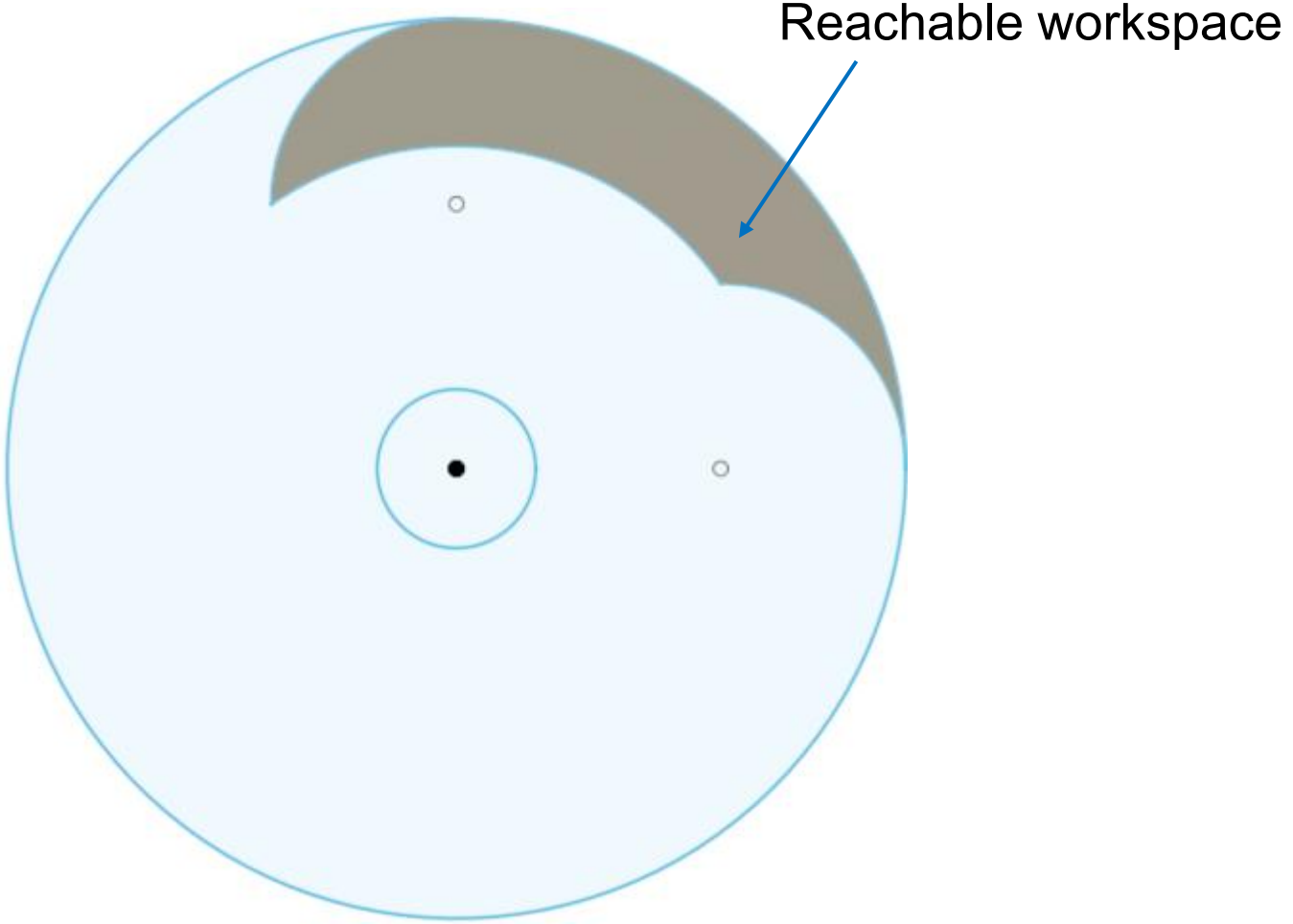
Dexterous workspace is an empty set.

Reachable workspace

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Workspace

$L_1 \neq L_2$. Say $L_1 = 1$ m, $L_2 = 0.7$ m.
$q_1 \in [0, \pi/2]$
$q_2 \in [0, \pi/2]$

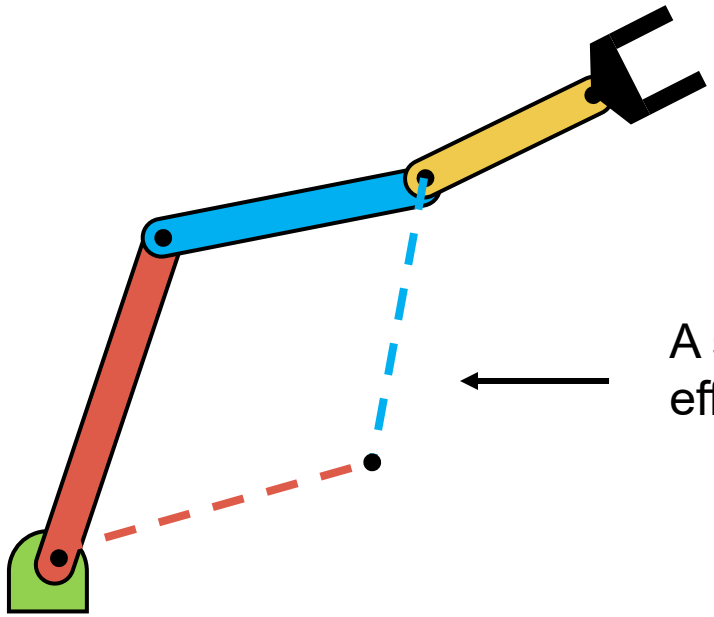


Dexterous workspace is an empty set.

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Multiple solutions to the IK problem may exist

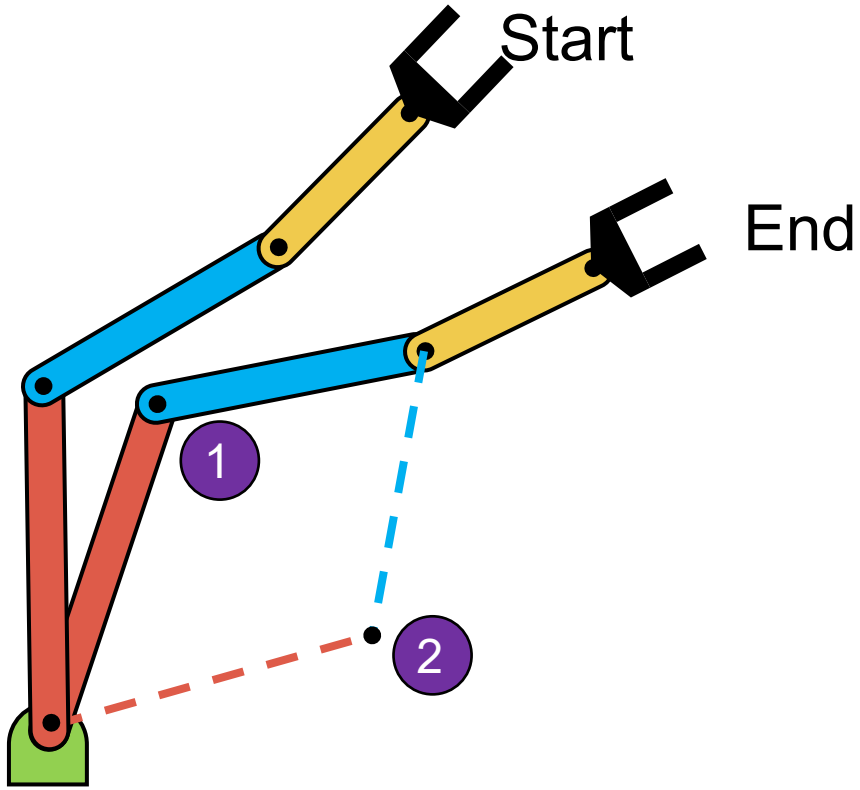
Consider this planar manipulator



A second configuration of the arm that places the end effector in the same position and orientation

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Choice of solution depends on the context



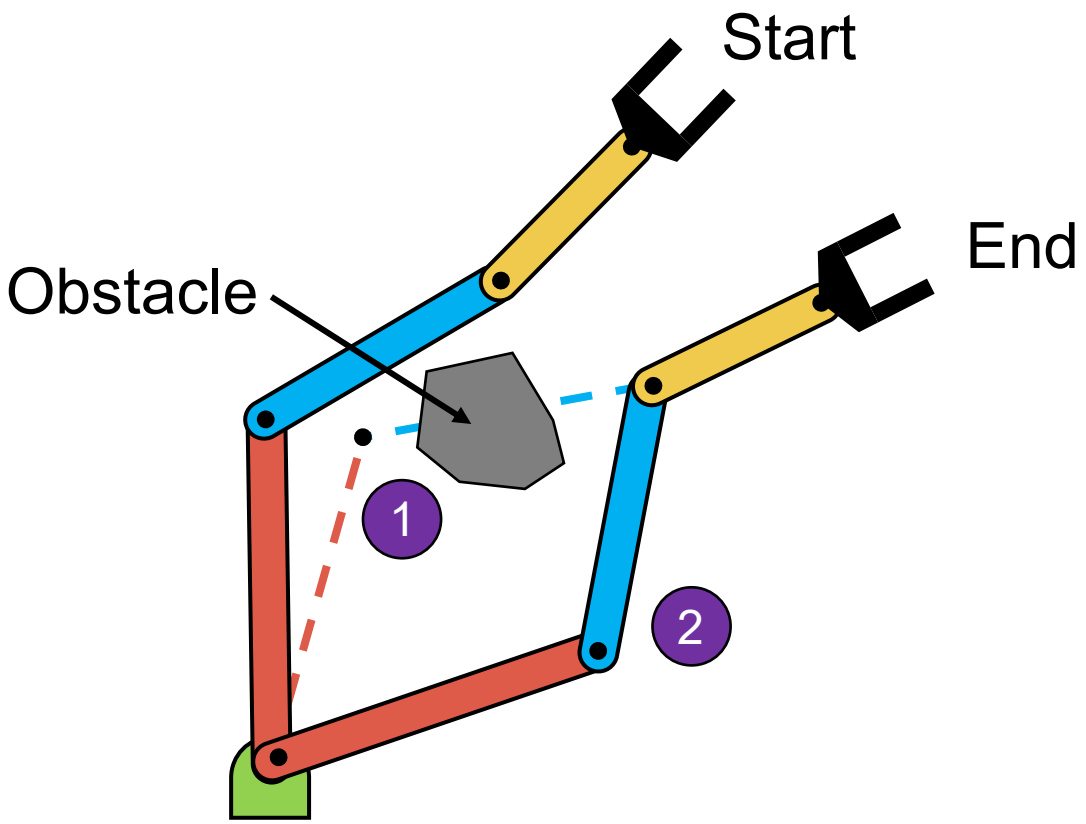
The configuration in solution 1 is “closer” to the initial configuration that solution 2.

Closeness can be defined in many ways:

- Absolute change in joint angles
- Weighted change in joint angles
- Number of joints to move
- The types of joints being moved

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Choice of solution depends on the context



In the presence of an obstacle, solution 2 is the only one that is practically possible!

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How is the IK problem solved?

- Closed-form solutions:
  - Based on analytical expressions or the solution of a polynomial.
  - Non iterative calculations suffice to arrive at a solution.
  - Typically rely on algebraic or geometric approaches.
- Numerical solutions:
  - Depend on iterative numerical techniques.
  - Slower than closed form solutions.
  - Work well for serial chains with many links.

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Closed-form / analytic solutions

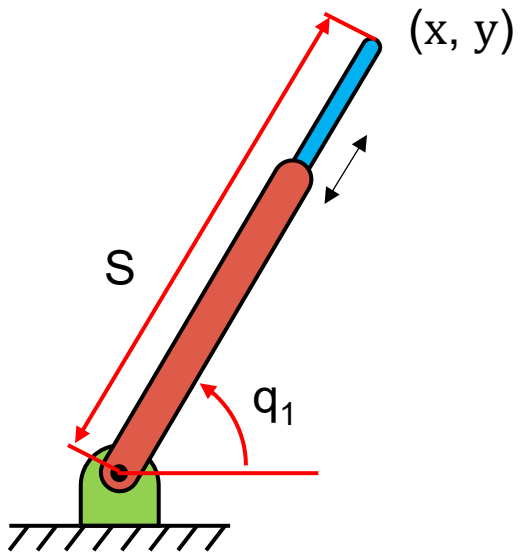
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The typical analytic approach to IK

- Create a model of the robot's links and joints
- Express forward kinematics Mathematically
- Using geometric or algebraic relations, attempt to solve for joint parameters
- Explicitly shows all the solutions – multiple configurations
- Solutions are compact and fast to execute

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Simple example – Planar RP manipulator



Given: x, y

Solve for: S, q_1

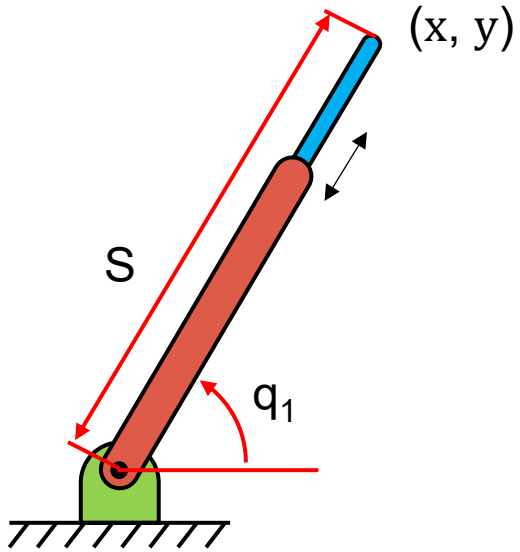
Solution:

$$q_1 = ?$$

$$S = ?$$

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Simple example – Planar RP manipulator



Given: x, y

Solve for: S, q_1

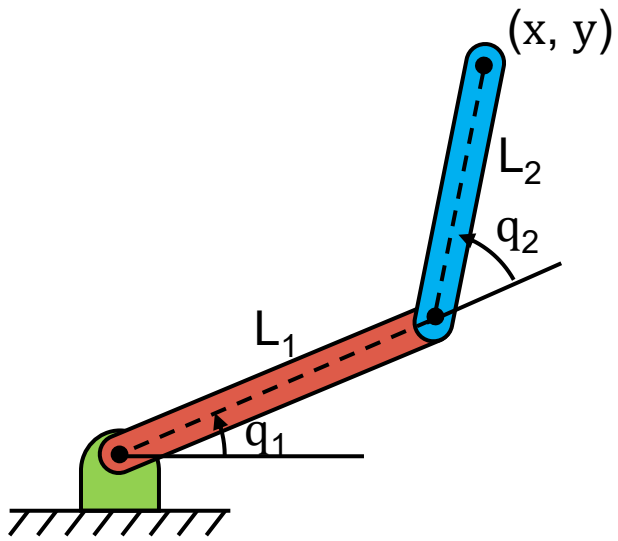
Solution:

$$q_1 = \tan^{-1}\left(\frac{y}{x}\right)$$

$$S = \sqrt{x^2 + y^2}$$

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Planar RR Robot



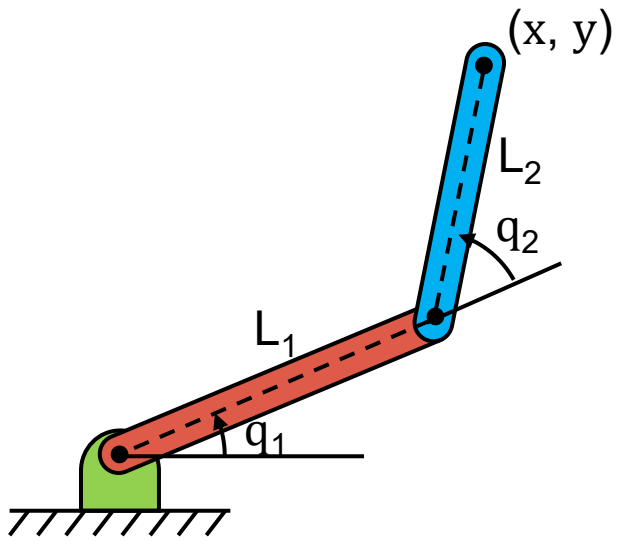
The IK problem:

Given: x, y, L_1, L_2

Solve for: q_1, q_2

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Planar RR Robot – Algebraic solution



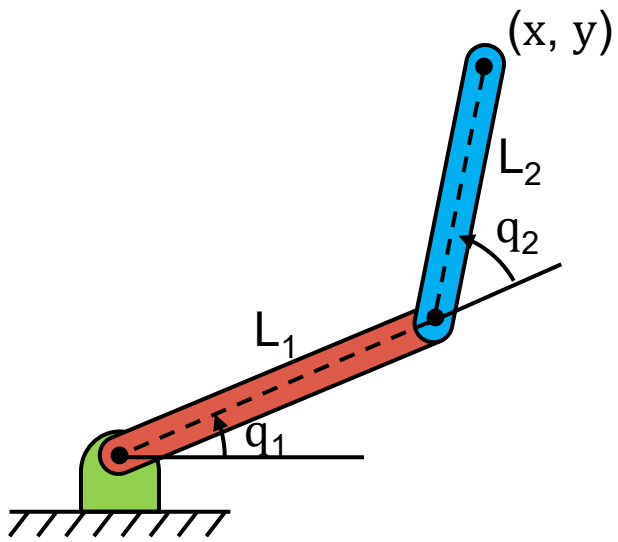
Forward Kinematics (considering only position):

$$x = L_1 \cos(q_1) + L_2 \cos(q_1 + q_2)$$

$$y = L_1 \sin(q_1) + L_2 \sin(q_1 + q_2)$$

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Planar RR Robot – Algebraic solution

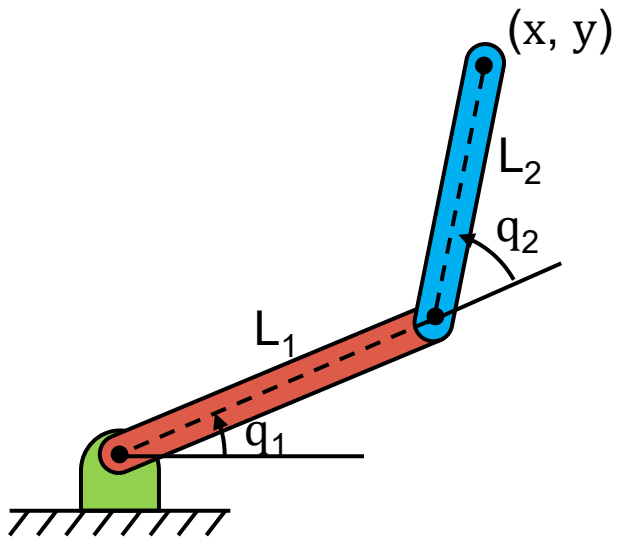


$$x = L_1 \cos(q_1) + L_2 \cos(q_1 + q_2)$$

$$y = L_1 \sin(q_1) + L_2 \sin(q_1 + q_2)$$

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Planar RR Robot – Solve for one joint angle



$$x = L_1 \cos(q_1) + L_2 \cos(q_1 + q_2)$$

$$y = L_1 \sin(q_1) + L_2 \sin(q_1 + q_2)$$

Squaring and adding,

$$x^2 + y^2 = L_1^2 + L_2^2 + 2L_1L_2 \cos(q_2)$$

$$\cos(q_2) = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}$$

$$\sin(q_2) = \pm\sqrt{1 - \cos(q_2)^2}$$

There are two solutions for q_2, given by $q_2 = \text{atan2}(\sin(q_2), \cos(q_2))$

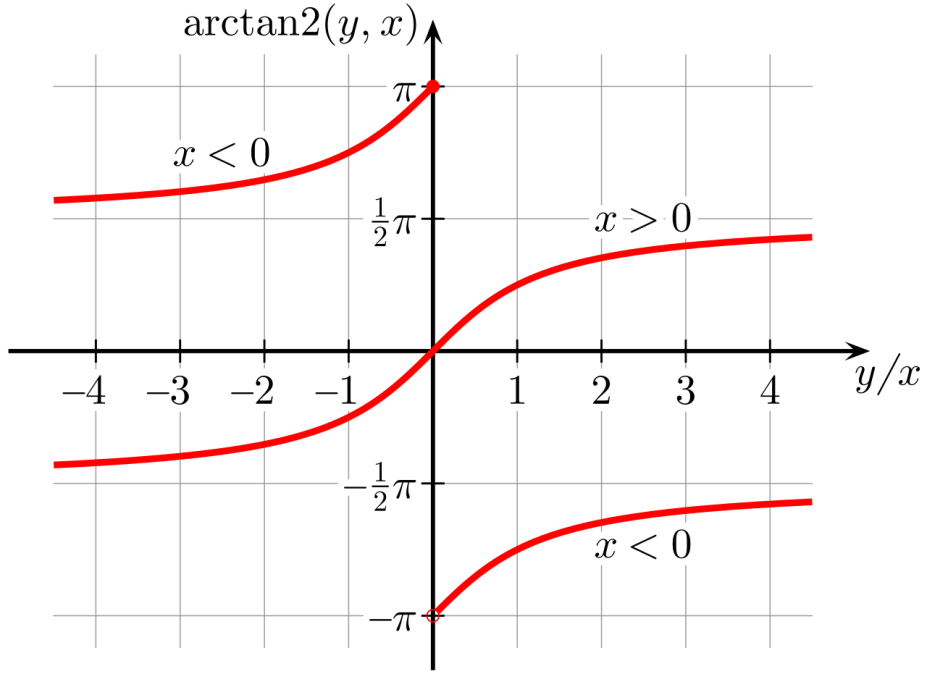
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A quick note about atan2(y,x)

- It is a version of $\tan^{-1}\left(\frac{y}{x}\right)$ that considers the sign of x and y. Takes two arguments.

$$atan2(y,x) = \begin{cases} \tan^{-1}\left(\frac{y}{x}\right) & \text{if } x > 0, \\ \tan^{-1}\left(\frac{y}{x}\right) + \pi & \text{if } x < 0 \text{ and } y \geq 0, \\ \tan^{-1}\left(\frac{y}{x}\right) - \pi & \text{if } x < 0 \text{ and } y < 0, \\ \frac{\pi}{2} & \text{if } x = 0 \text{ and } y > 0, \\ -\frac{\pi}{2} & \text{if } x = 0 \text{ and } y < 0, \\ undefined & \text{if } x = 0 \text{ and } y = 0 \end{cases}$$

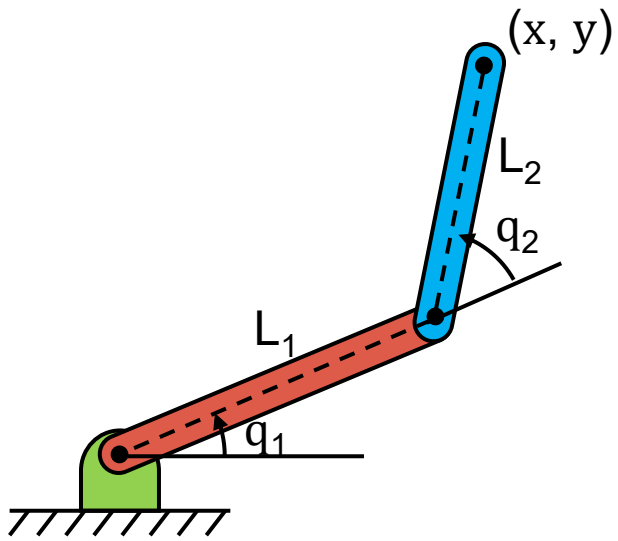
Range = [- π, π]



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Planar RR Robot – Now solve for the other joint angle



$$x = L_1 \cos(q_1) + L_2 \cos(q_1 + q_2)$$

$$y = L_1 \sin(q_1) + L_2 \sin(q_1 + q_2)$$

Use compound angle formulae

$$x = (L_1 + L_2 \cos(q_2)) \cos(q_1) - (L_2 \sin(q_2)) \sin(q_1)$$

$$y = (L_2 \sin(q_2)) \cos(q_1) + (L_1 + L_2 \cos(q_2)) \sin(q_1)$$

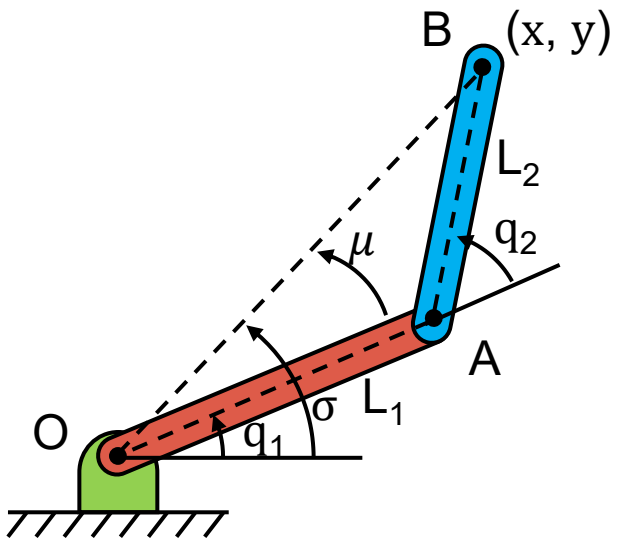
General solution for an equation of the type $A \cos\varphi + B \sin\varphi = C$ is

$$\varphi = \tan^{-1}\left(\frac{C}{\sqrt{A^2 + B^2 - C^2}}\right) - \tan^{-1}\left(\frac{A}{B}\right)$$

$$q_1 = \tan^{-1}\left(\frac{y}{x}\right) - \tan^{-1}\left(\frac{L_2 \sin(q_2)}{L_1 + L_2 \cos(q_2)}\right)$$

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Planar RR Robot – Geometric Solution



$$x^2 + y^2 = L_1^2 + L_2^2 - 2L_1L_2\cos(\pi - q_2)$$ Cosine rule in ΔOAB

$$\cos q_2 = \left(\frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}\right)$$ ← Two solutions from $\tan^{-1}(\quad)$

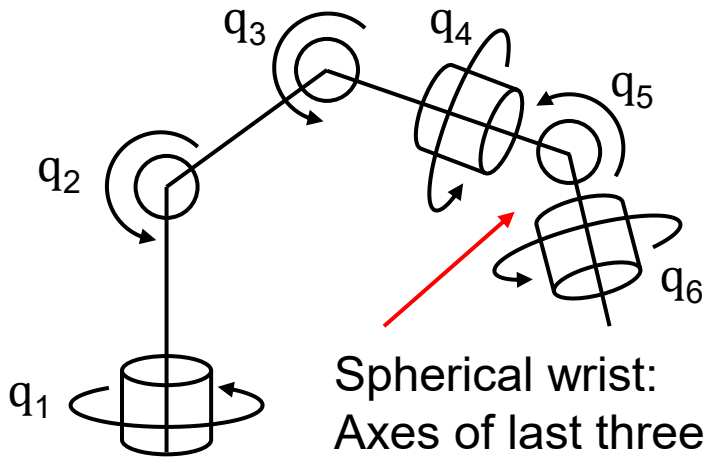
$$\frac{\sin(\mu)}{L_2} = \frac{\sin(\pi - q_2)}{\sqrt{x^2 + y^2}} = \frac{\sin(q_2)}{\sqrt{x^2 + y^2}}$$ Sine rule in ΔOAB

$$\sigma = \text{atan2}(y, x)$$

$$q_1 = \sigma - \mu$$ ← Two solutions since q_2 has two solutions

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6DoF 3D manipulator



Spherical wrist:
Axes of last three joints intersect in a point

The IK problem:

Given: A desired pose of the end effector

$$ {}^{0}T_{E} = \begin{bmatrix} \mathrm{R}_{3\times3} & \mathrm{T}_{3\times1} \\ 0_{1\times3} & 1 \end{bmatrix} $$

Solve for: $q_1, q_2, q_3, q_4, q_5, q_6$

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IK of 6DoF 3D manipulator

- The desired pose (given as a transformation matrix), encodes six degrees of freedom
  - This information can be represented by 6 numbers.
  - Since we only have six joints at our disposal, there are no redundant solutions.
- For this manipulator, the problem can be decomposed into two
  - Position component – dictated by the first three joints
  - Orientation component – dictated by the last three joints

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First, work out the position of the wrist

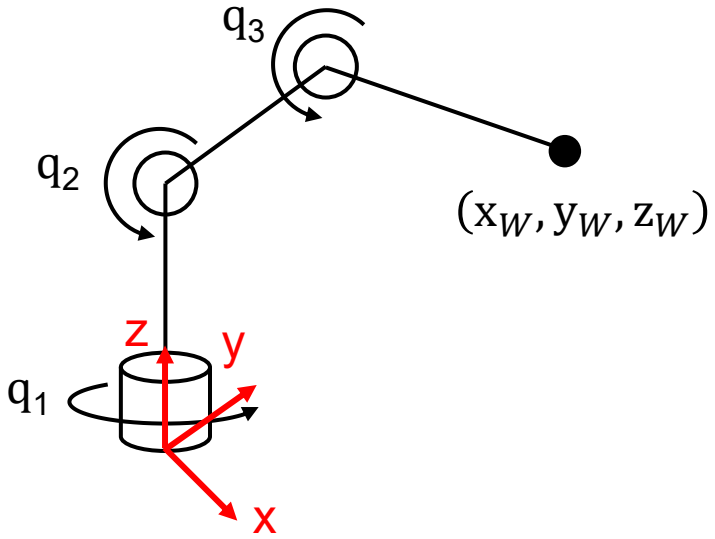
- In the spherical wrist, all three joints are rotating about the same point.

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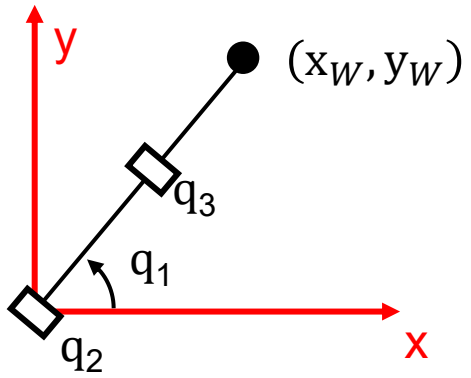
Then, solve for the first 3 joints

Step 1: solve for q_1

$$q_1 = \text{atan2}(y_W, x_W)$$



Looking down from above

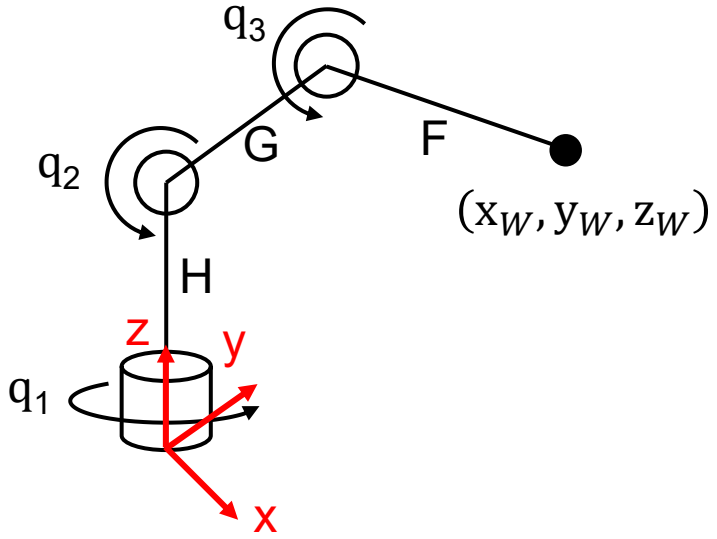


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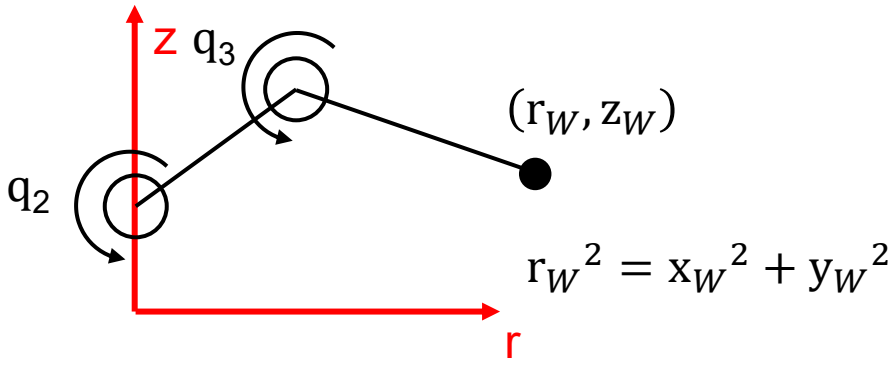
Solving for the first 3 joints – step 2

Step 2: solve for q_3

$$\cos(\pi - q_3) = \frac{r_W^2 + (z_W - H)^2 - F^2 - G^2}{2FG}$$



Look at the plane in which the arm is located



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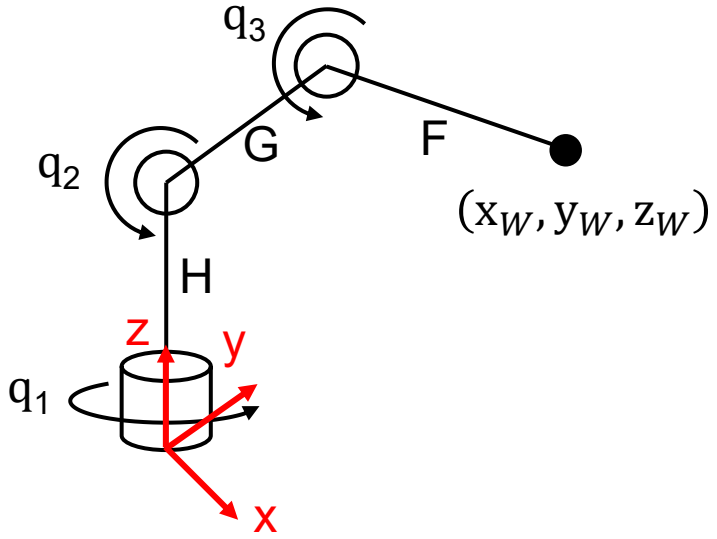
Solving for the first 3 joints – step 3

Step 3: solve for q_2

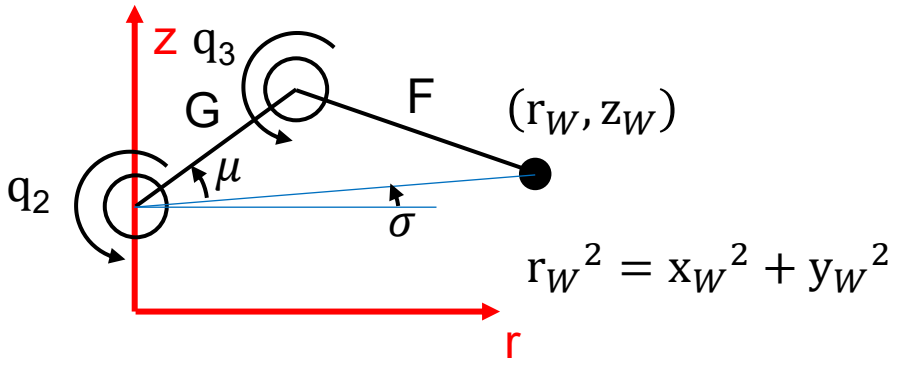
$$q_2 = \sigma + \mu$$

$$\sigma = \text{atan2}(z_W - H, r_W)$$

$$\mu = \text{atan2}(F\sin(q_3), G + F\cos(q_3))$$



Look at the plane in which the arm is located



$$r_W^2 = x_W^2 + y_W^2$$

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Finally, compute the rotations in the wrist

- ${}^{3}T_{6}$ is composed of three rotations.
- This is like the Euler angles we studied in the previous lecture.
- Use the algebra from compound rotations to determine the three joint angles.

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What if a robot doesn’t have a spherical wrist?

What if it is too complicated to solve the IK problem analytically?

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Numerical solutions

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The general idea of numerical solutions

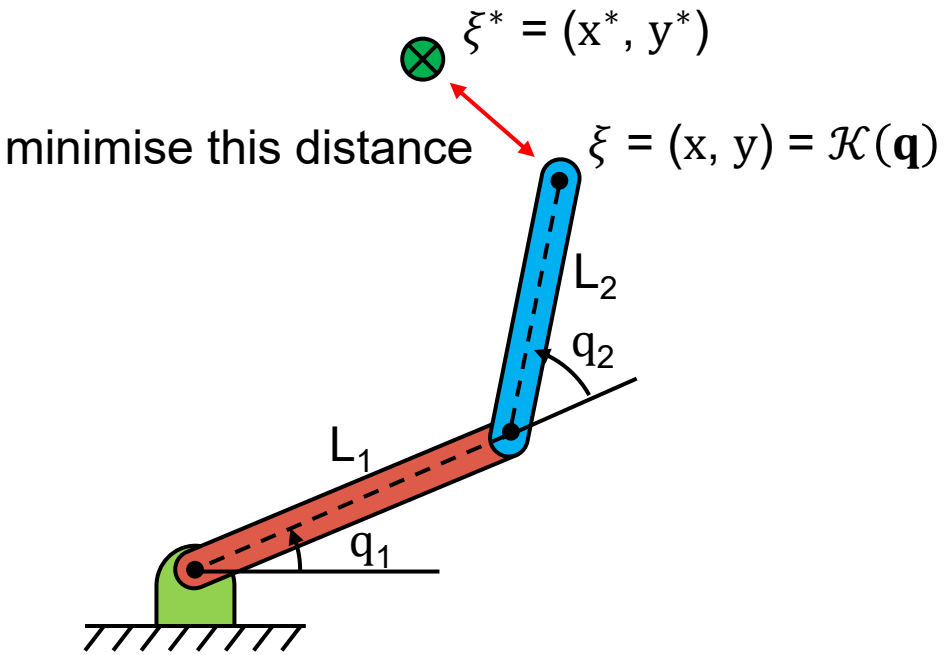
- We can always determine the forward kinematics of a robot: $\xi_n = \mathcal{K}(\mathrm{q})$.
- Say, we want to reach a goal pose of ξ^*.
- Adjust the joint parameters q until $\xi \rightarrow \xi^*$.
- The resulting set of joint parameters q^* is the solution to the IK problem.

This idea can be posed as an optimisation problem:

$$\mathrm{q}^* = \arg\min_{\mathrm{q}} |\mathcal{K}(\mathrm{q}) \ominus \xi^*|$$

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The goal potential can be defined in many ways



Squared Euclidian distance is one common definition of $G(q)$

$$G(\mathbf{q}) = |\mathcal{K}(\mathbf{q}) \ominus \xi^*| = (x^* - x)^2 + (y^* - y)^2$$

If considering only position and no orientation

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Things to consider

- Is it possible to always achieve $G(\mathbf{q}) = 0$?
- Will the solution always be found, if it exists?
- Is the solution unique?

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Things to consider

- Is it possible to always achieve $G(\mathbf{q}) = 0$?

  No. ξ^* could be from the unreachable workspace.

- Will the solution always be found, if it exists?

  No. The solver may hit a local optimum.

- Is the solution unique?

  No. The solution depends on initial condition and the redundancy in the robot.

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One simple numerical approach

Newton-Raphson method:

- We want to solve the equation $G(\mathrm{q}) = 0$ for a differentiable function $G: \mathbb{R} \to \mathbb{R}$
- Start with an initial guess of the solution: q^0

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One simple numerical approach

Newton-Raphson method:

- We want to solve the equation $G(\mathrm{q}) = 0$ for a differentiable function $G: \mathbb{R} \to \mathbb{R}$
- Start with an initial guess of the solution: q^0
- Write the Taylor series expansion for $G(\mathrm{q})$ at q^0 and truncate it to first order

$$G(\mathrm{q}) = G(\mathrm{q}^0) + \frac{\partial G}{\partial \mathrm{q}}(\mathrm{q}^0)(\mathrm{q} - \mathrm{q}^0) + h.o.t$$

Ignore higher order terms

- Set $G(\mathrm{q}) = 0$, and solve for q

$$\mathrm{q} = \mathrm{q}^0 - \left(\frac{\partial G}{\partial \mathrm{q}}(\mathrm{q}^0)\right)^{-1} G(\mathrm{q}^0)$$

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Newton-Raphson method continued

$$\mathrm{q} = \mathrm{q}^0 - \left(\frac{\partial G}{\partial \mathrm{q}}(\mathrm{q}^0)\right)^{-1} G(\mathrm{q}^0)$$

Use this q as a new guess and repeat. An iterative expression can be written down as follows.

$$\mathrm{q}^{k+1} = \mathrm{q}^k - \left(\frac{\partial G}{\partial \mathrm{q}}(\mathrm{q}^k)\right)^{-1} G(\mathrm{q}^k)$$

Repeat the above until a suitable stopping criterion is reached.

$$\left|\frac{G(\mathrm{q}^{k+1}) - G(\mathrm{q}^k)}{G(\mathrm{q}^k)}\right| \leq \epsilon$$ ← Some arbitrarily chosen threshold

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Newton-Raphson applied to IK

Consider only position for now

- Forward kinematics gives us $\xi_n = \mathcal{K}(\mathrm{q})$

- Consider the part that gives the position of the end effector

  $\boldsymbol{x} = \mathcal{F}(\mathrm{q})$ where $\boldsymbol{x} \in \mathbb{R}^m$ and $\mathrm{q} \in \mathbb{S}^1 \times \mathbb{S}^1 \cdots \times \mathbb{S}^1$, n times

- The inverse kinematics problem (applied to only position) is:

  when given a desired end effector position $\boldsymbol{x}^*$

  find q^* such that $G(\mathrm{q}^*) = |\mathcal{F}(\mathrm{q}^*) \ominus \boldsymbol{x}^*| = 0$

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Newton-Raphson applied to IK

Consider only position for now

- The inverse kinematics problem (applied to only position) is to

$$\text{find } \mathrm{q}^* \text{ such that } G(\mathrm{q}^*) = |\mathcal{F}(\mathrm{q}^*) \ominus \boldsymbol{x}^*| = 0$$

- Let us start with an initial guess q^0

- We saw earlier that the update equation is

$$\mathrm{q}^{k+1} = \mathrm{q}^k - \left(\frac{\partial G}{\partial \mathrm{q}}(\mathrm{q}^k)\right)^{-1} G(\mathrm{q}^k)$$

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Newton-Raphson applied to IK

Consider only position for now

$$\mathrm{q}^{k+1} = \mathrm{q}^{k} - \left(\frac{\partial G}{\partial \mathrm{q}}\left(\mathrm{q}^{k}\right)\right)^{-1} G\left(\mathrm{q}^{k}\right)$$

The quantity $\frac{\partial G}{\partial \mathrm{q}}\left(\mathrm{q}^{k}\right) = \frac{\partial \mathcal{F}}{\partial \mathrm{q}}\left(\mathrm{q}^{k}\right)$ is called the coordinate Jacobian, $J\left(\mathrm{q}^{k}\right) \in \mathbb{R}^{m \times n}$

If $J\left(\mathrm{q}^{k}\right)$ is square and invertible, iteratively update q until $G\left(\mathrm{q}^{k+1}\right)$ converges to 0.

In other words, we find a q^{k+1} where $\mathcal{F}\left(\mathrm{q}^{k+1}\right)$ is close to our desired $\boldsymbol{x}^{*}$

This q^{k+1} is the solution to the inverse kinematics problem: q^{*}

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What if the Jacobian is not square?

- If J is not square, then it is not invertible
- We can solve (approximately) for $\Delta \mathrm{q}$ by using the Moore-Penrose pseudo inverse: $J^{\dagger}$
- When J is of full rank,

$J^{\dagger} = J^{\intercal}(JJ^{\intercal})^{-1}$ if J is fat, i.e., $n > m$. There can be infinite solutions to $\Delta \mathrm{q}$.

$J^{\dagger} = (J^{\intercal}J)^{-1}J^{\intercal}$ if J is tall, i.e., $n < m$. There is no exact solution to $\Delta \mathrm{q}$.

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Algorithm

- Given: a desired position $\boldsymbol{x}^* \in \mathbb{R}^m$ and an initial guess $\mathrm{q}^0 \in \mathbb{R}^n$
- Initialise $i = 0$
- Set $e = \boldsymbol{x}^* - \mathcal{F}(\mathrm{q}^i)$
- While $|e| > \epsilon$:

  Set $\mathrm{q}^{i+1} = \mathrm{q}^i + J^{\dagger}(\mathrm{q}^i)e$

  Increment i

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What if we want a desired pose?

- Instead of just a desired position $\mathbf{x}^*$, we want a desired pose ξ^*
- $\xi^* \in SE(3)$ whereas $\mathbf{x}^* \in \mathbb{R}^m$

- Homework: Read chapter 6.2.2 of Modern Robotics by Lynch and Park

- Hint: you need to use the twist vector and body Jacobian

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Closed-form vs. Numerical solutions

- Closed form solutions are potentially faster than an iterative solution.
- The benefit of a closed-form solution is that a unique solution to all manipulator positions can be determined a priori.
- Finding a closed-form solution becomes increasingly difficult as the number of joints increases.
- Numerical methods are more suited to large chains.
- They help determine a close enough solution when an exact one does not exist.

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Summary

- Inverse kinematics is the problem of determining how to get the robot to a desired pose.
- Sometimes an analytical solution exists and can be arrived at by using algebra or geometry.
- When it doesn’t exist, a numerical method can be used.

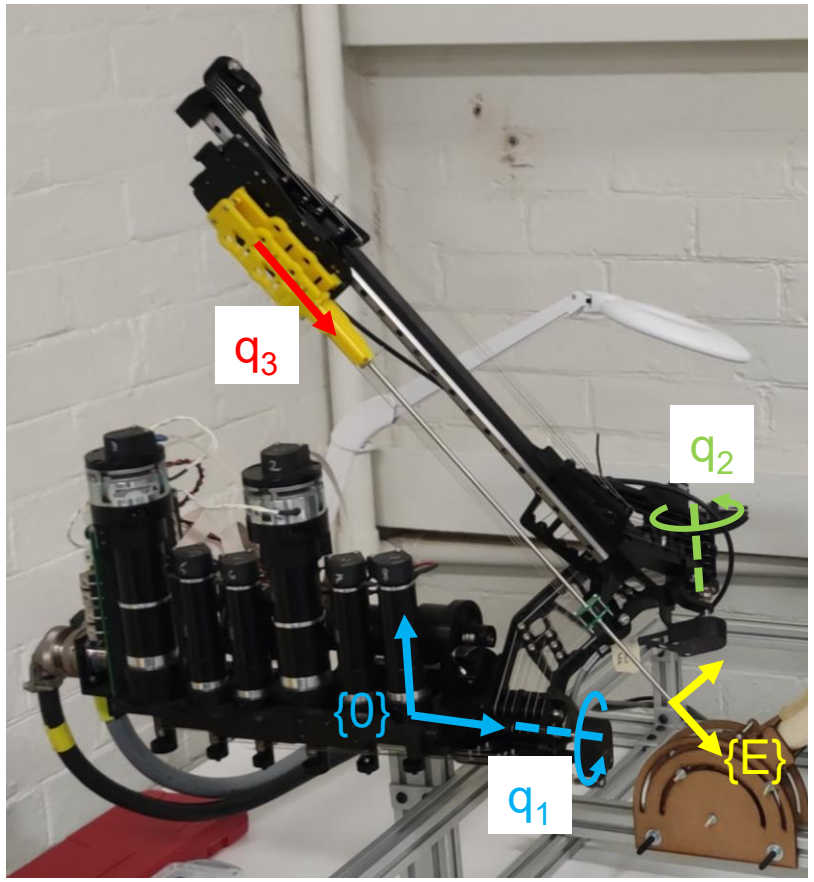
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Summary

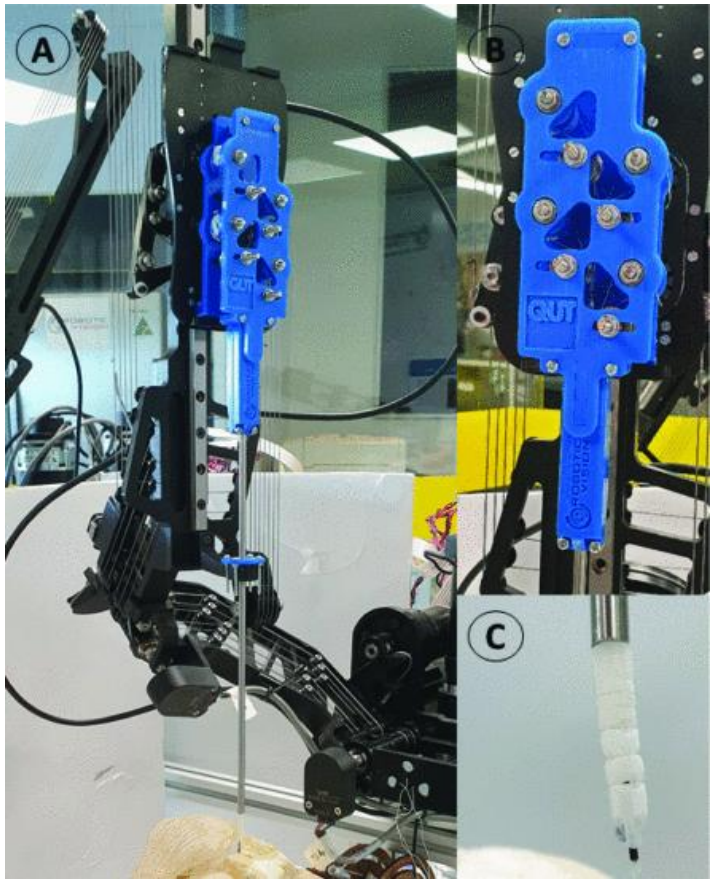
- All systems with revolute and prismatic joints having a total of 6 DoF in a single series chain are solvable.
- However, the general solution may not have an analytic form.
- A sufficient condition that a manipulator with 6 DoF has a closed-form solution is that three neighbouring joint axes intersect at a point.

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Go beyond



The Raven II surgical robot at QUT



Andrew Razjigaev, et al

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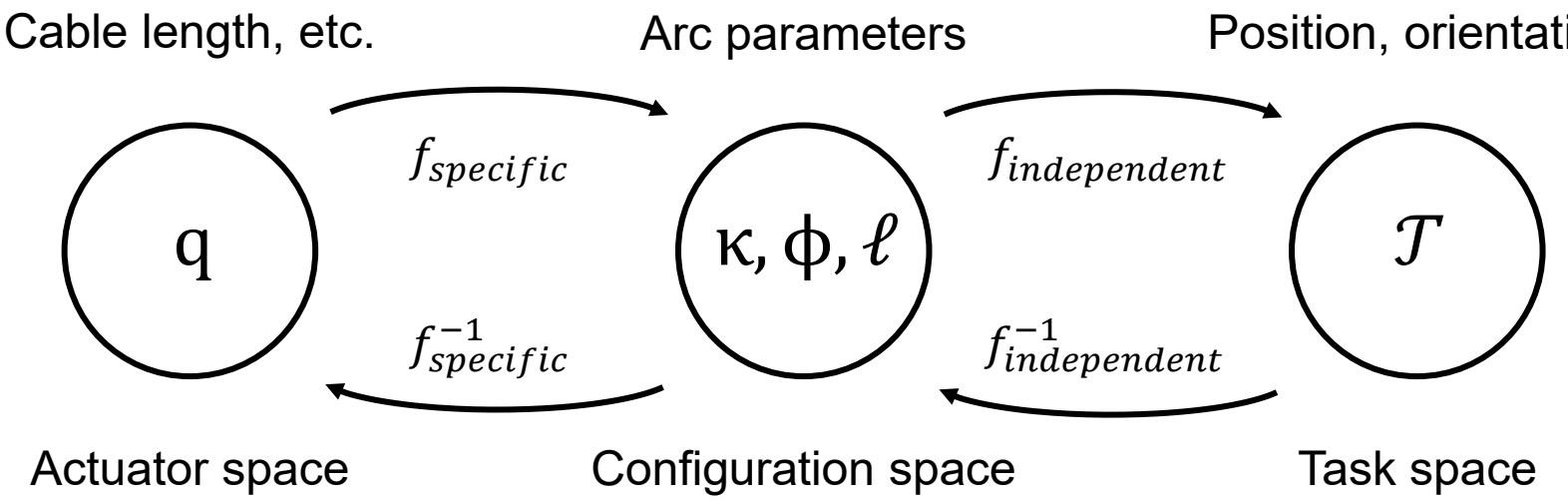
Go beyond



Vasquez J, Hackaday, 2016

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In continuum robots, configuration space is different from actuator space

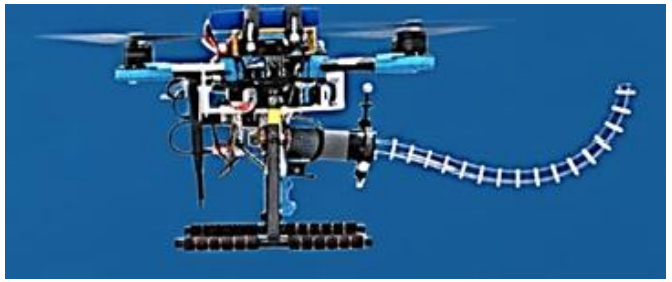


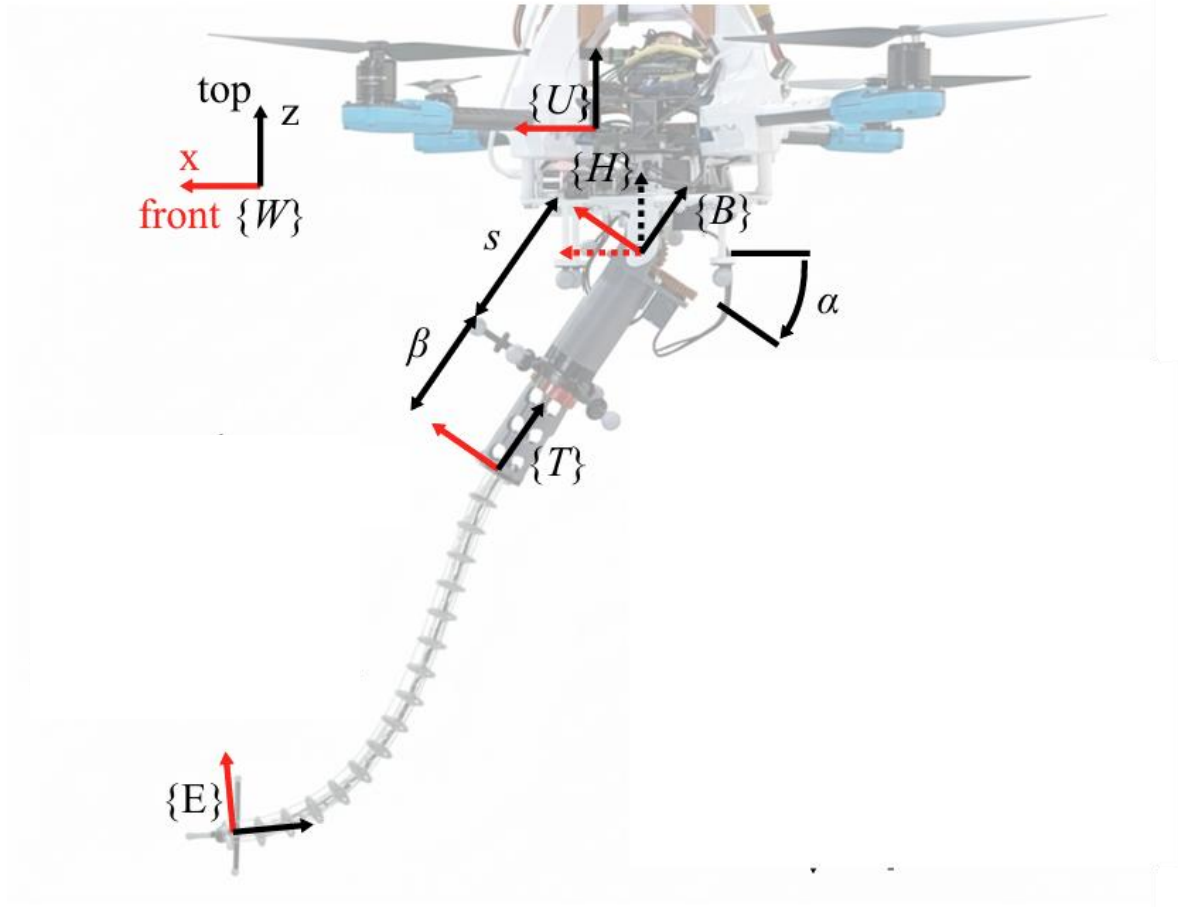
Webster III RJ, Jones BA. Design and kinematic modeling of constant curvature continuum robots: A review. The International Journal of Robotics Research. 2010 Nov;29(13):1661-83.

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Go beyond







Anuraj Uthayasooriyan, et al. ICRA 2026
Tilt-X: Enabling Compliant Aerial Manipulation through a Tiltable-Extensible Continuum Manipulator

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Next week

- We will discuss the kinematics (both forward and inverse) of parallel link manipulators.



ABB FlexPicker

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Questions?

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